TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Chapter 1

Introduction

This document is a community-made reference guide for Falcon BMS 4.38.1, focused on practical use of three specific HOTAS controls: the Target Management Switch (TMS), the Display Management Switch (DMS), and the Countermeasures Management Switch (CMS). Although developed under Falcon BMS 4.38.1, the fundamental behavior of these switches has remained consistent since at least Falcon BMS 4.36, making this guide applicable to virtually any Falcon BMS user.

The goal of this guide is to reorganize information scattered across the Dash-1, Dash-34, and the BMS Training Manual into mode-based tables and concise explanations. Virtual pilots can quickly find what each switch press does in a given context, answering questions such as: *"In this radar mode, what does TMS Up do?"* or *"How do I cycle through MFD formats with the DMS?"*

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Nothing in this document should ever be used for real-world operations, training, or procedures.

1.1 Scope and purpose

This guide focuses exclusively on three HOTAS switches: the Target Management Switch (TMS), the Display Management Switch (DMS), and the Countermeasures Management Switch (CMS). Although other controls exist on the F-16 throttle and stick—such as the Communication Switch, the Dogfight/MRM Override, and the RDR Cursor Enable control—they could be mentioned only when essential to understanding the behavior and context of TMS, DMS, and CMS.

This is not a comprehensive HOTAS or avionics manual. Instead, it is a usage guide organized by operational context, with emphasis on practical tables that show what each switch input does in specific flight modes, sensor configurations, and weapon employment scenarios. The guide bridges information scattered across official documentation and training missions, making it immediately accessible to pilots who need quick, authoritative answers.

The guide assumes knowledge of basic F-16 operation and familiarity with master modes (NAV, A-A and A-G). It does not replace the Dash-34 or Training Manual; rather, it complements them by organizing TMS, DMS, and CMS behavior into searchable tables and diagrams with cross-references back to official sources and practical training missions where each behavior can be practiced.

1.2 Sources and references

This guide is grounded in the following primary Falcon BMS documents:

1. **TO BMS 1F-16CMAM-34-1-1** (Dash-34, Change 4.38) — Avionics and Nonnuclear Weapons Delivery Flight Manual. Primary source for HOTAS behavior, SOI logic, and weapons employment.

2. **BMS Training Manual 4.38.1** (October 2025) — Training missions and learning objectives. Used for practical cross-references and recommended training progression.

3. **TO BMS 1F-16CMAM-1** (Dash-1) — F-16 Aircraft Systems, Normal and Abnormal Procedures. Reference for general aircraft systems context.

4. **BMS User Manual 4.38** — BMS user interface, DTC configuration, and simulation setup.

5. **MCH 11-F16 Vol 5** (May 1996) — F-16 Combat Aircraft Fundamentals, Surface Attack and Weapons Delivery. Supplementary tactical reference.

All technical claims in this guide are cross-referenced to these sources. Where specific sections are cited, the notation Dash-34 § X.Y.Z refers to Dash-34 section X.Y.Z, and TRN NN refers to BMS Training Mission NN.

1.3 Document structure and how to read it

1.3.1 Chapter overview

This guide is organized into seven chapters. Chapters 2 through 6 follow a consistent structure combining narrative explanation with detailed reference tables. The table below summarizes the focus of each chapter:

Table 1.1: Guide Chapter Overview

Chapter	Focus
C2	HOTAS fundamentals: Sensor of Interest (SOI), short vs. long press timing, master modes, and switch overview
C3	TMS — Target Management Switch: targeting, designation, and track management
C4	DMS — Display Management Switch: SOI selection and MFD format cycling
C5	CMS — Countermeasures Management Switch: CMDS and ECM control
C6	Training references: recommended progression and practical mission flows
C7	Diagrams

Chapter 7 — Visual Quick Reference: Unlike the preceding chapters, which combine narrative explanation with detailed tables, Chapter 7 serves as a standalone quick reference. It consolidates all TMS, DMS, and CMS switch functions documented in Chapters 3–5 into schematic diagrams organized by operational context, without descriptive text. This format is designed for rapid in-flight consultation, allowing pilots to quickly verify switch behavior without reading through explanatory sections.

1.3.2 Table format

Chapters 3, 4, and 5 use a standardized seven-column table format called `hotastable` to document switch behavior. Each row describes a single switch action in a specific operational context:

Table 1.2: HOTAS Table Column Definitions

Column	Description
Mode	Operational context: master mode, sensor state, or weapon configuration
Dir.	Switch direction: Up, Down, Left, or Right
Act.	Actuation type: Short press or Long press
Function	Brief name of the function performed
Effect / Nuance	Detailed explanation of system behavior, constraints, and tactical considerations
Dash34	Cross-reference to Dash-34 section(s)
Train.	Recommended BMS training mission(s) for hands-on practice

To find information in a HOTAS table: (1) identify the master mode and sensor context from the section title; (2) locate the relevant Mode within the table; (3) find the Direction and Action; (4) read the Effect and consult the Dash34 or Training reference for further detail.

1.3.3 Reference conventions

This guide uses consistent notation for cross-references to official documentation:

- **Dash-34 § X.Y.Z** — Reference to Dash-34 (TO BMS 1F-16CMAM-34-1-1) section X.Y.Z
- **TRN NN** — Reference to BMS Training Mission number NN
- **MCH 11-F16 Vol 5** — Referenced by full title when cited
- **User Manual § X.Y** — Reference to BMS User Manual section X.Y

These conventions appear throughout the guide in narrative text and in the Dash34 and Train. columns of HOTAS tables.

1.4 Version, authorship, and AI assistance

1.4.1 Document version and status

Table 1.3: Document Status

Item	Value
Document Version	0.3.2.1+20260129
Falcon BMS Version	4.38.1 (Update 1)
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Chapters with Content	3/7
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1.4.2 Authorship and AI assistance

This guide was created by a member of the Falcon BMS community with structured assistance from AI language models:

- **Perplexity AI** — Research organization, initial structuring, and early content generation (January 2026)
- **Claude (Anthropic)** — Co-editor: critical review, governance framework, content development, and ongoing revision (January–February 2026)

The human author identified scope, validated all content against official Falcon BMS documentation, made all organizational and editorial decisions, and bears full responsibility for the guide's accuracy and presentation. AI tools were used for research organization, cross-referencing, structural analysis, and text generation—not for defining technical correctness.

1.4.3 Canonical source

The canonical source for this guide, including all governance documents, version history, and contribution guidelines, is the project repository:

https://github.com/carlos-nader/tms-dms-cms-usage-guide

This repository contains the authoritative version of all project files. Translations, adaptations, and derivative works should reference this repository as the original source.

1.4.4 License

1.4.5 Disclaimer

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